

KARMUKILAN

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EDUCATION

National Institute of Science Education and Research (NISER)

Aug 2023 – Present

Bhubaneswar, India · Integrated BS-MS, Mathematics · CGPA: 8.58/10 · 4th Year (expected graduation 2028)

Relevant Coursework: Probability Theory, Measure Theory, Real Analysis I & II, Metric Space Theory, Differential Equations, Complex Analysis, Calculus of Several Variables, Graph Theory, Abstract Algebra (Group, Ring & Field Theory), Mathematical Methods in Physics

CS Academy, Coimbatore, India — CBSE Grade 12, 90% (2023) · Maharishi Vidya Mandir Sr. Sec. School, Kumbakonam, India — CBSE Grade 10, 94% (2021)

RESEARCH EXPERIENCE

Competitive Brownian Bug Model

May 2026 – Present

IFISC (CSIC–UIB), Palma de Mallorca, Spain · Supervisor: Dr. Emilio Hernández-García

- Modelling competitive population dynamics as a stochastic birth–death process; derived closed-form expressions for the first and second moments of the population size directly from the system’s master equation in the high-diffusion limit.
- Validating the analytical predictions against direct numerical simulation of the underlying stochastic process.

Mean-Field Kuramoto Model

Winter 2024

SASTRA Deemed University, Thanjavur, India · Supervisor: Dr. V. K. Chandrasekhar

- Proved linear stability of the incoherent (disordered) phase of the mean-field Kuramoto model in the thermodynamic limit.
- Computed the complex order parameter against coupling strength for finite oscillator ensembles, characterising the synchronisation transition.

Non-linear Dynamics & Mathematical Modelling

Summer 2024

IISER Thiruvananthapuram, India · Supervisor: Dr. Chandrakala Meena

- Studied bifurcation theory, perturbation methods, and phase-portrait analysis for non-linear dynamical systems near fixed points; numerically reproduced soliton solutions of the non-linear Schrödinger equation with a cubic potential.

Functional Analysis

Summer 2025

IIT Guwahati, India · Supervisor: Dr. Arup Chattopadhyay

- Characterised bounded linear operators on Banach spaces via the Hahn–Banach Theorem, the Uniform Boundedness Principle, and the Open Mapping Theorem; explored topological structure on operator spaces.

Integration Theory & Fixed-Point Theory

Summer 2025

IIT Tirupati, India · Supervisor: Dr. Rajesh

- Constructed a variational structure on the partition space of a compact interval via a scalar functional, and proved Brouwer’s Fixed-Point Theorem using Sperner’s Lemma.

SKILLS

Quantitative: Stochastic Processes & Master Equations, Probability Theory, Measure Theory, Dynamical Systems & Bifurcation Theory, Partial Differential Equations, Functional Analysis, Mathematical Modelling

Technical: Python (numerical simulation & data analysis), LaTeX, MS Excel / Word / PowerPoint

HONORS & SCHOLARSHIPS

DISHA–DAE Scholarship — Department of Atomic Energy, Government of India (merit-based)